

Energy Metric Interview

Audio Transcript

Hello, everybody. My name is Maria Noguer and I'm a doctor in physics and climate change. I work at the Institute for Environmental Analytics, and I'm the climate programme manager there at the University of Reading.

The team at the IEA have been involved in developing Energy Metric out of a project that was funded by the UK Space Agency. Energy Metric offers a solution to support renewable energy planning and help regions and nations with their net zero emissions. Energy Metric supports energy planners, both in public and private sectors, utility companies, energy consultants and renewable energy project developers in general.

We are working with some energy companies in Colombia. These energy companies are in charge of distribution and also generation of electricity, and quite a few of them, they need to prepare management plans for how the energy system will evolve in the face of climate variability and change. And they need to plan to horizons of around 10 to 20 years. The key things that these energy companies are facing, is it comes from two places if you like. One is damage to the physical assets that they've got due to increase in extreme events. And I can talk a little bit more about that in a minute. But the second one is an increase in the demand of electricity, which is just not triggered by an increase in population, which is obviously the population increase, the demand increase, but it's also by climate change because an increase in temperature will increase this demand for electricity.

For example, in terms of extreme events in South America, they have been hit lately badly by landslides and, for example, also by wildfires. Wildfires are amplified by fire weather conditions, as we call fire weather, which is high winds, high air temperatures, low relative humidity, low rainfall. All these are some of the changes that we are seeing and we expect it to be worse with climate change in some parts of South America. So basically, these are the challenges that these companies are facing: physical damage and demand on electricity.

Climate change is expected to increase the energy demand to support adaptation measures such as, for example, increasing in cooling and perhaps also water supply for agriculture and humans because with climate change, we're expecting to see parts of the world being drier so irrigation might increase as well in those parts of the world.

All these things are energy intensive, things that require energy. If you use renewable energy for those sort of energy intensive activities, you will at the same time adapt but also reduce your greenhouse gases. So renewable energy is kind of like for both. Obviously, it's a mitigation measure because you reduce your greenhouse emissions, but it's also an adaptation measure. Sometimes people forget about the adaptation side of renewable energy, of installing renewable energy. An increase in temperature of one degree will trigger a different demand for energy.

But in terms of generation, in the next 10 or 30 or 20 years, the generation is dominated by the weather variability and not by climate change in the sense of that it's going to increase the generation or anything like that. The important thing is generation is dominated by weather variability. What we do is we create these 10 years of weather data, historical weather, the recent 10 years, which capture that variability. We use these 10 years for each of the scenario years that we plan. What we're doing

at Energy Metric is what we call 'Renewable Energy Scenario Planning'. We explore different configurations of technology, solar or wind, to meet a certain penetration of renewable energy by a certain year.

For example, imagine that a country needs to, by 2030, they want to have a 30% penetration of renewable energy. A scenario in Energy Metric is a configuration of assets that the planner will put in different places in the country, plus a demand. And that's the important thing. The demand is what varies. For 2025 we will have a demand of X megawatts that they need. For 2030 that demand will increase. But the weather that drives the simulation is exactly the same. The weather variability, the 10 years that we've produced is used for the 2025 scenario and for the 2030 scenario. The difference is the demand and the configuration of assets that the planner introduces.

The future energy system needs to cope with higher and higher proportions of variable renewable energy, which depends on weather. And as we know, the wind does not always blow, the Sun does not always shine so we have this information in our hands to be able to plan for better renewable energy. The viability of generation of wind and solar depends on understanding the risks and impacts of variable weather. The key weather data relevant for wind and solar production, and wave as well, is obviously wind speed incoming short-wave radiation, temperature, and with wave amplitude. The wave is less developed, but Energy Metric still have got a way to produce that as well. These are the key variables.

How do we put that together in a dataset? Okay, so we use weather observations. We use satellite data products, and we use modelling techniques. The whole point is to create a complete, gridded dataset over the region of interest. How do we do that? Our starting point is something called 'reanalysis data'. This is taken from the Copernicus Climate Data Store, which is freely available, and this dataset is global and it's got spatial resolution of around 30 by 30 kilometers. So the whole globe is divided into squares of 30 by 30 kilometers and there is information on weather and climate for every 30 by 30 kilometers. It's historical information back to 40 to 50 years.

What happens with this dataset is that it doesn't capture details of wind flow around complex terrains or cloud formations in the way that you need for renewable energy production. What we do then is we take this reanalysis data and we downscale it using a regional weather model which is tailored and bespoke to the region of interest. When we say downscaling, we use a technique which is a dynamical downscaling, better than a statistical method, because what happens with dynamical downscaling is that it's based upon the explicit representation of the physical principles. We use the laws of thermodynamics to see how the sun comes out and the clouds are formed, fluid mechanics to see how the wind flows around the complex terrain. This is a numerical model based on physical principles.

What's the result of this? The result is then multi-year historical weather data to ensure we capture the year-by-year variability and the extremes of this local weather and the local climate.

In this way, we ensure that our solar and wind resource assessments are reliable and representative, that's very important, of the climate at the site of interest.

What's the next step after that? These weather datasets that we have created then are the input to Energy Metric, which is our software platform. So what is it the platform does? The platform then just transforms weather into power. It allows the user, the client, to access these enhanced datasets and use them to provide these renewable energy resource estimates for investing and planning purposes.

What we do is enhance these freely available datasets into two components. One is the production of this bespoke, multi-year, high-resolution weather data, which we have a technique to do that through this regional weather model, and this dataset can then be purchased separately for the clients to use in their own application so the data can be purchased separately. But also, the second component of Energy Metric is the access to the software platform that converts weather into power so that the clients can explore different potential scenarios and see what's the best pathways for them.